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C&O Group – Smart Multi-Division Event Management & Client Booking Platform

Final Year Project Synopsis

Submitted by

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Project Overview

Project Title:
C&O Group – Smart Multi-Division Event Management & Client Booking Platform

Project Category:
Startup-Based Project

Domain:
Event Technology (EventTech) | B2B & B2G Event Management

Organization:
C&O Group LLP

- *Thread & Knot* – Wedding & Social Events
- *Prism Planner* – Corporate, MICE & Government Events

Project Description:
This project focuses on designing and developing a centralized, web-based platform that digitizes end-to-end event management workflows. The system integrates client interaction, booking management, vendor coordination, invoicing, and operational tracking under a single platform, specifically tailored for startup-driven, compliance-oriented event organizations.

Key Focus:
Unified operations, compliance-first execution, scalability, and startup-ready MVP architecture.

Specific Objectives

- To design and develop a **centralized, web-based event management platform** supporting multiple event divisions under a single brand.
- To enable **structured client interaction** for browsing services, viewing portfolios, comparing packages, and submitting inquiries or bookings.
- To provide a **secure admin dashboard** for managing bookings, clients, event packages, vendors, and operational workflows.
- To automate **key operational processes** such as booking approvals, invoice generation, email notifications, and QR-based event passes.
- To ensure **compliance, transparency, and documentation readiness** for corporate and government-oriented events.
- To implement a **scalable, startup-ready MVP architecture** suitable for asset-light operations and future expansion.

Key Features

- **Multi-Division Event Management**
  Supports multiple business divisions such as wedding/social events and corporate/MICE events under a single platform.

- **Client Booking & Inquiry System**
  Enables clients to explore services, compare event packages, submit inquiries, and initiate bookings digitally.

- **Centralized Admin Dashboard**
  Provides administrators with full control over bookings, clients, event packages, vendors, and operational data.

- **Vendor Management Module**
  Allows assignment and tracking of vendors including décor, catering, photography, and logistics services.

- **Automated Invoice Generation**
  Generates professional, downloadable invoices in PDF format for approved bookings.

- **QR-Based Event Passes**
  Issues QR codes for event access and verification, improving operational efficiency and security.

- **Email Notifications & Status Updates**
  Sends automated confirmations and real-time updates for inquiries and booking status changes.

Project Usecases & Scope

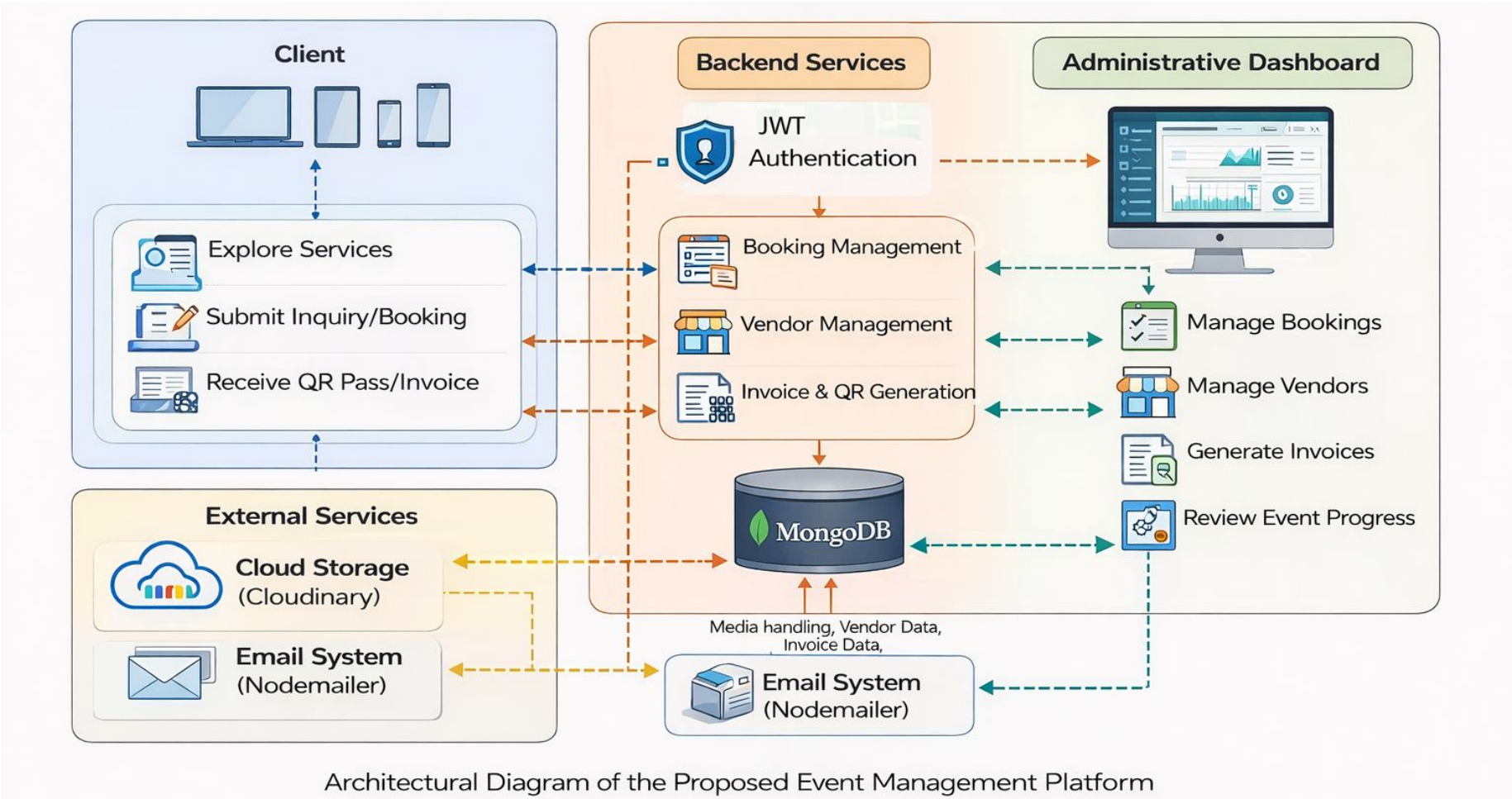
Use Cases:

- Corporate organizations planning conferences, product launches, and business events
- Government bodies managing summits, public programs, and official events
- Event management teams handling multi-event operations under one platform
- Clients tracking booking status, invoices, and event details digitally

Project Scope:

- Web-based platform for end-to-end event management
- Client-side booking, inquiry, and service exploration
- Admin-side control for approvals, vendors, invoices, and event lifecycle
- QR-based passes and automated communication
- Designed as a **startup-ready MVP** for scalability and compliance

Prototype/Project Flow diagram/Architecture



Architectural Diagram of the Proposed Event Management Platform

Data & Resources

Data Sources:

- Client-provided data such as inquiries, booking details, and event preferences
- Admin-managed data including event packages, vendor information, and pricing
- System-generated data such as booking status logs, invoices, and QR pass records

Databases & Software Resources:

- **MongoDB:** Used to store structured and unstructured application data
- **Node.js & Express.js:** Backend processing and API management
- **JWT Authentication:** Secure access control for users and administrators

Cloud & Digital Resources:

- **Cloudinary:** Storage and management of event images and portfolio media
- **PDF & QR Code Libraries:** Generation of invoices and event passes
- **Email Services:** Automated notifications and confirmations

Methodology

Approach:

- The system is developed using a **modular full-stack architecture** to manage client interactions and administrative operations in a unified platform.
- Client requests (inquiries and bookings) are processed through a secure backend, where business logic handles validation, approval workflows, and status updates.
- Administrative operations such as vendor assignment, package management, and invoice generation are controlled through a centralized dashboard.
- Automation is applied to repetitive processes to reduce manual effort and improve operational efficiency.

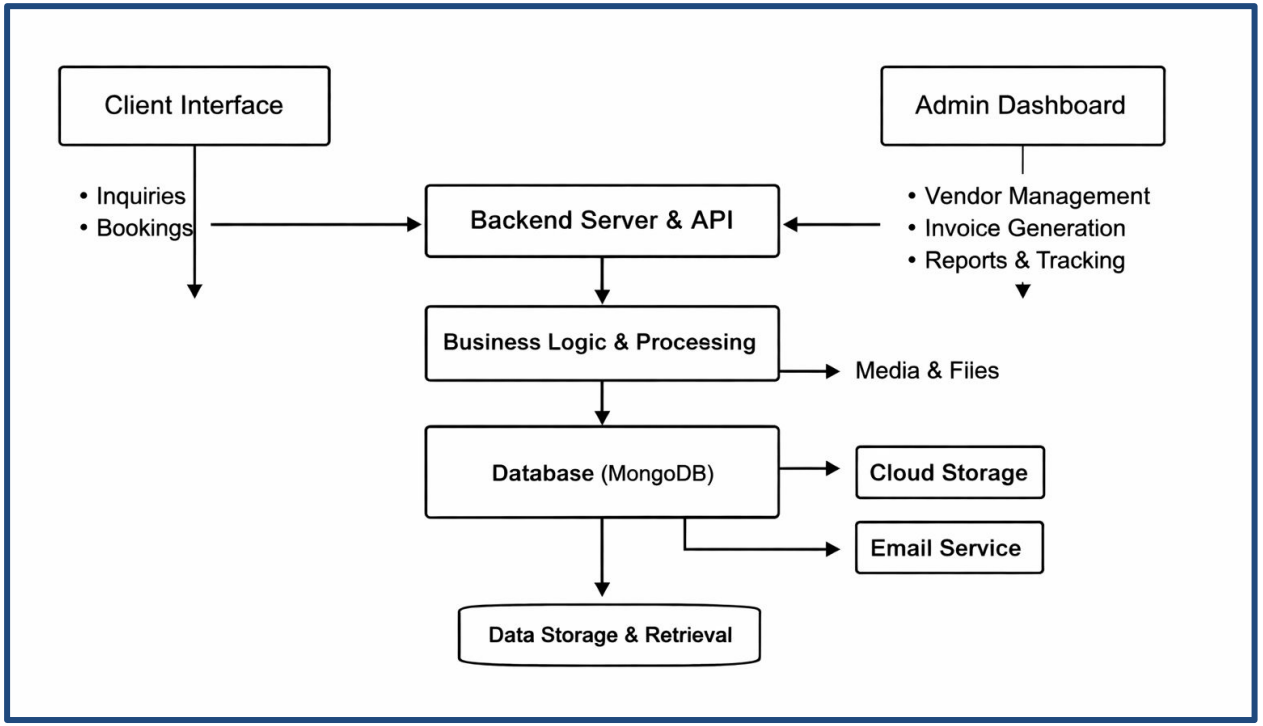
Tools, Technologies & Frameworks:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Authentication:** JWT-based role authorization
- **Cloud Services:** Cloudinary for media storage
- **Utilities:** PDF generation for invoices, QR code generation for event passes
- **Communication:** Email notification service

Methodology

System Flow (Block Diagram):

- Client Interface → Backend Server → Database
- Admin Dashboard ↔ Backend Server
- Backend ↔ Cloud Storage & Email System



Expected Results & Impact

Expected Results:

- A fully functional, web-based event management platform for C&O Group LLP
- Streamlined client booking and inquiry process
- Centralized admin system for managing packages, vendors, and bookings
- Automated invoice generation and QR-based event passes
- Secure data handling with role-based access control

Impact:

- Improves operational efficiency and reduces manual workload
- Enhances transparency and documentation for corporate and government events
- Supports startup-oriented, asset-light business operations
- Provides a scalable digital foundation for future expansion
- Demonstrates practical application of full-stack development in a real-world domain

Project Timeline

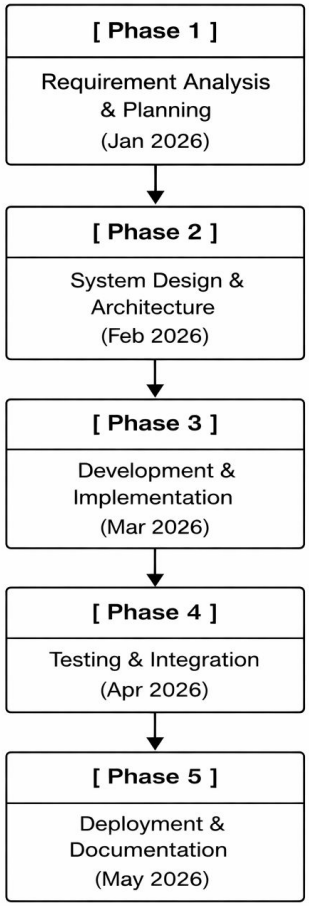


Figure: Project Timeline Flow Diagram

THANK YOU